

Embodying Affect: A Review of Emotion Modeling, Detection, and Expression in Immersive Virtual Agents

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Abstract. Emotions play a critical role in human cognition and interaction, influencing perception, decision-making, and social behavior. As Virtual Reality (VR) evolves into a platform for complex social interaction, the need for virtual agents capable of modeling, detecting, and expressing emotions becomes paramount. This short paper reviews the current state of affective computing in immersive environments. We analyze computational models of emotion, focusing on the shift from theoretical appraisal models to interoperable software frameworks. We examine the detection of user states through multimodal physiological signals and deep learning fusion. Finally, we explore how agents express affect, highlighting the impact of avatar realism on user embodiment and empathy. We conclude that the future of immersive agents relies on closed-loop systems that integrate physiological detection with biologically inspired cognitive architectures.

1 Introduction

Emotions are essential to rational decision-making, perception, and learning, playing a critical role in human behavior [2]. Virtual Reality (VR) has emerged as a powerful tool for simulating complex real-world situations in controlled laboratory conditions, offering high levels of presence and interactivity [2]. As VR environments become more realistic, the development of "Interactive Virtual Reality Environments" that can interpret and react to human emotions is crucial for applications in training, therapy, and education [8].

To create emotionally competent virtual agents, research must address three pillars: (1) computational models that define how an agent processes emotion; (2) detection mechanisms that accurately perceive the user's state using implicit measures; and (3) expression mechanisms that allow agents to respond appropriately [9]. This paper reviews recent literature to propose an integrative view of these components.

2 Computational Models of Emotion

To endow an agent with affect, a theoretical framework is required. Historically, research has been divided between *Appraisal Theories* and *Dimensional Theories*.

2.1 Appraisal and Dimensional Frameworks

Appraisal theory is a dominant paradigm in Artificial Intelligence (AI) because it links emotion to cognition [5]. It posits that emotion arises from an individual's judgment of the relationship between events and their beliefs, desires, and intentions [5]. Commonly used models include the OCC model (Ortony, Clore, and Collins) and Lazarus's theory [5].

In contrast, dimensional theories map affective states onto continuous axes. The most common is the Circumplex Model of Affect (CMA), which utilizes Valence (pleasantness), Arousal (activation), and Dominance (control) [2]. While appraisal models are often used for derivation, dimensional models are frequently employed for the recognition and continuous representation of affect in computational systems [3].

2.2 Interoperability in Cognitive Architectures

A significant challenge in computational modeling is the integration of affective components into broader cognitive architectures. Contemporary models often interact in an ad-hoc manner, making it difficult to incorporate new cognitive components developed by third parties [4].

Osuna et al. propose an interoperable framework to address this, utilizing "syntactic interoperability" for data exchange and "semantic interoperability" via ontologies to interpret cognitive data [4]. This approach allows cognitive information—such as memory or attention—to bias affective components, treating emotion as a modulator of cognition rather than an isolated module [4].

3 Multimodal Emotion Detection

For an agent to respond emotionally, it must detect the user's state. While self-reports are used, they can disrupt immersion; thus, the field is shifting toward implicit measures using physiological signals [2].

3.1 Physiological Signals and Deep Learning

Common implicit measures include Heart Rate Variability (HRV) and Electrodermal Activity (EDA), which reflect Autonomic Nervous System (ANS) activity [2]. However, single modalities often lack the resolution to distinguish complex emotional states. Recent research emphasizes multimodal fusion, combining signals such as Eye-Tracking (ET) and Galvanic Skin Response (GSR) [3].

Deep learning architectures have demonstrated superior performance in this domain. Qu et al. proposed the Multi-Modal Temporal Emotion Detector (MMTED), which integrates Convolutional Neural Networks (CNNs) and Transformers to process GSR and eye-movement data [3]. This fusion approach achieved recognition accuracy as high as 89.27%, significantly outperforming traditional machine learning methods [3].

3.2 Multi-Domain Sensing

Improving detection also involves capturing signals from multiple anatomical domains. Indrasiri et al. introduced a framework collecting data from the Head (VR headset eye-tracking), Trunk (ECG vest), and Periphery (wrist EDA) [1]. Their architecture, utilizing Multi-Scale Attention mechanisms, showed that fusing these domains provides a more holistic view of emotional states, yielding higher accuracy in detecting valence and arousal compared to single-domain models [1].

4 Expression and Interaction

Once modeled and detected, the agent must express a reaction. This output is often mediated through avatars or environmental changes.

4.1 Avatar Expressiveness and Realism

Avatars facilitate social interaction and embodiment in VR [7]. High expressiveness in avatars is generally believed to enhance engagement. However, Ponce et al. found that while user personality (specifically extraversion) correlates with a sense of embodiment, increased facial expressiveness in an avatar did not significantly alter the user's reported emotional state or anxiety during a speech task [7].

Conversely, the synchronization of verbal and non-verbal cues is critical. Du et al. found that in VR, users are highly sensitive to mismatches between gesture and voice [6]. While VR enhances the perception of natural gestures, it widens the perceptual gap for synthetic ones. Interestingly, a "semi-synthetic" approach (e.g., natural voice with synthetic gestures) can still elicit strong empathetic responses, suggesting perfect realism is not always necessary if at least one modality is high-fidelity [6].

4.2 Affective Visualization

Beyond avatars, VR systems can visualize the user's internal state to provide biofeedback. Pinilla et al. review "Affective Visualization," where the system represents the user's stress or relaxation visually (e.g., changing colors or game elements) [9]. This creates a closed loop that enables therapeutic interventions, allowing users to learn how to modulate their neural activity or physiological states [9].

5 Conclusion

The literature reveals a convergence toward closed-loop affective systems in VR. We are moving away from static stimulus-response models toward dynamic architectures where detection uses deep learning to fuse multimodal data [3], modeling relies on interoperable frameworks [4], and expression balances naturalism with synthesis to enhance empathy [6]. Future research must focus on validating these systems in wilder, more interactive contexts to bridge the gap between theoretical models and practical application.

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